

Scenario 04: GreenCharge

Company Overview

GreenCharge operates a network of electric vehicle (EV) charging stations across Western Europe. Founded in 2020, they've grown to 2,400 charging points in 8 countries. The company focuses on "destination charging" —locations where drivers spend time anyway (shopping centers, hotels, office parks).

Business Model:

- Per-kWh charging fees (€0.35-0.55/kWh depending on location and speed)
- Monthly subscription: €12.99/month for 15% discount on all charging
- B2B partnerships: Revenue share with host locations (shopping centers, hotels)

Current Metrics:

- 2,400 charging points across 340 locations
- 185,000 registered app users
- 28,000 monthly subscribers
- 1.2M charging sessions/month
- Average session: 45 kWh, €19.80 revenue
- Station utilization: 18% average (varies widely by location)

The Situation

GreenCharge is facing a critical decision. They have €15M allocated for expansion, but the board is divided on strategy:

Option A: Expand into 3 new countries (Poland, Czech Republic, Portugal)

Option B: Densify the network in existing countries (add more stations in proven locations)

Option C: Focus on improving utilization at existing underperforming stations

The CEO, Lars Bergström, favors Option A for growth narrative. The COO, Maria Santos, prefers Option B, arguing that network density drives user loyalty. The CFO, Thomas Mueller, is pushing

Option C, claiming that 40% of stations are operating below breakeven utilization.

Recent data shows concerning patterns:

- 35% of stations have <10% utilization
- Users in areas with more GreenCharge stations use the network 2.5x more
- Subscriber churn increased from 4% to 7% monthly after a competitor entered the German market

Available Data

You have access to:

1. **Stations table:** station_id, location_id, country, city, location_type (mall/hotel/office/other), charger_speed (kW), installation_date, monthly_rent_cost
2. **Sessions table:** session_id, station_id, user_id, start_time, end_time, kwh_delivered, revenue, payment_method
3. **Users table:** user_id, signup_date, country, car_model, subscription_status, subscription_start
4. **Locations table:** location_id, partner_name, revenue_share_pct, foot_traffic_estimate, parking_spaces
5. **Competitor stations:** latitude, longitude, operator, charger_speed (scraped from public APIs, updated monthly)

Data quality notes:

- Foot traffic estimates are provided by partners and may be inflated
- Competitor data has 2-3 week lag and may be incomplete
- Session data is clean but some sessions have missing user_id (guest charging)
- Subscription cancellation reasons are not captured

Key Stakeholders

- **Lars Bergström (CEO):** Wants aggressive growth story for next fundraiser. Believes "be everywhere" is the winning strategy.
- **Maria Santos (COO):** Focused on operational efficiency. Believes density in existing markets creates competitive moat.

- **Thomas Mueller (CFO):** Concerned about cash burn. Wants to fix underperforming stations before building new ones.
- **Country Managers (8):** Each wants expansion in their territory. May resist data showing their stations underperform.
- **Board of Directors:** Split between growth and efficiency. Expecting a data-driven recommendation.

Constraints

- Expansion analysis must be ready for board meeting in 4 weeks
- Cannot get user-level data from competitors (only station locations)
- Country-specific regulations affect pricing and partnership structures
- €15M budget is fixed—cannot recommend "all of the above"

The Ask

The board wants a data-driven recommendation for the €15M expansion budget. Specifically:

1. What drives station profitability? Which station types/locations should we prioritize?
2. Does network density actually improve user retention and usage? What's the evidence?
3. Where should the €15M go: new countries, densification, or fixing existing stations?

Thomas has privately noted: "Lars is emotionally attached to the expansion narrative. We need objective analysis."

Your deliverable: Create a complete Analytics Project Brief for this analysis.